

XIN BAI

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EDUCATION

University of Utah

Salt Lake City, UT

B.S. in Electrical Computer Engineering

12/2025

TECHNICAL SKILLS

SolidWorks	Python/C++	PLC	SLAM	OrCAD	Sensor
EDA:Pspice/LTspice	Digital System Design 3D	MATLAB	PCB	Ardupilot(APM)	PX4
Python	COMSOL	AutoCAD	Linux	Gazebo	ROS

PROFESSIONAL EXPERIENCE

Quicktron Automation

Project Engineer

07/2026

- **Supported end-to-end AGV/AMR project deployment**, including site infrastructure setup, system commissioning, UAT execution, production launch, and post-deployment maintenance to ensure stable warehouse automation operations.
- **Performed advanced AGV troubleshooting across software, network, electrical, and hardware layers**, using Final Shell to access Linux/ROS environments, retrieve and analyze system logs, verify network connectivity and device communication, and isolate root causes between PLC, safety LiDAR, controllers, communication modules, and AGV hardware.
- **Validated RCS and production-line integrations through API-based task dispatching and end-to-end workflow testing**, including transport-order creation, button-box functionality, station interaction, AGV task execution, and system status feedback during UAT and production support.
- **Configured and developed system nodes using LES Lite**, supporting workstation logic, equipment interfaces, task-flow configuration, and integration between RCS, AGVs, button boxes, and production-line equipment.
- **Developed strong hands-on knowledge of AGV architecture and field maintenance**, including safety LiDAR, PLC, CBox/DBox, motor controllers, drive motors, lifting mechanisms, batteries, sensors, communication networks, and related electrical components, enabling efficient fault diagnosis, component replacement, and recovery of production operations.

University Of Utah

Hardware & Software Engineering

01/2025

- Engineered a **robust UAV stabilization architecture** that improving controllability and safety by 35% in turbulent airflow conditions common to real-world urban and natural environments. Integrated GPS, IMU, and digital airspeed sensors (MATEKSYS ASPD-4525) with a Pixhawk flight controller.
- Implemented **end-to-end sensor integration** across GPS, IMU, and airspeed systems, improving state estimation fidelity and flight robustness.
- Designed and validated **real-time wind estimation and sensor-fusion algorithms** in MATLAB, directly informing adaptive control strategies.
- Delivered **low-level firmware and middleware integrations** within PX4 and ArduPilot, spanning build systems, cross-compilation, and custom sensor drivers for real-time flight control.

BYD

Facility Management Engineer

05/2023

- Improved **sensor positioning accuracy by 25%**, directly increasing marking-head precision and repeatability through calibration and hardware optimization.
- Performed **hardware-level inspections and diagnostics** on electrical and electromechanical systems supporting precision equipment, ensuring stable and compliant operating conditions for engineering hardware.
- Developed and executed **preventive maintenance plans** that improved 30% reliability of electrical and mechanical assets used in production and testing environments.
- Improved overall system throughput by **troubleshooting hardware issues, tuning system performance, and optimizing operational workflows**.

AECC Guizhou Liyang Aero-Engine Co.

Hardware & Software Engineer

08/2023

- Optimized **CNC/laser marking processes** for aerospace turbine blades, reducing QR code engraving deviation from 2–8 mils on ~80% of parts to ≤ 3 mils on all parts, significantly improving marking precision, consistency, and traceability compliance.
- Deployed **Linux-based control software updates** and integrated upgraded print-head hardware, enhancing machine performance with 15% higher efficiency and 30% improved reliability in production environments.
- Supported **on-site aerospace production operations**, working directly with CNC/laser marking equipment in factory environments to diagnose issues, calibrate hardware, and ensure continuous system availability under production constraints.